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PALEOMAGNETIC STUDY OF A VERTICAL SEQUENCE OF TRAPS FROM MOUNT PAVAGARH, GUJRAT, INDIA

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The paleomagnetism of 22 flows which range in composition from olivine basalts to rhyolite and in elevation from 450 ft. to 2680 ft., from Mount Pavagarh, situated in Gujrat, India, has been studied. The igneous activity represented by these flows belongs to the Deccan Trap Plateau Basalt Series. Sixteen flows ranging in elevation from 900 to 2680 ft. showed normal magnetization with upward inclination, consistent with the position of India in the southern hemisphere during the period of their eruption. Four flows from elevation 450 ft. to 730 ft. showed intermediate as well as discordant directions. The paleomagnetic results obtained from these flows have been correlated with those of Deccan traps from other areas. The amount of paleosecular variation represented by these flows has been estimated and compared with that from similar vertical sequences of traps studied from other areas. It has been found that the magnitude of paleosecular variation varied during the period of Deccan trap activity. An estimate of continental drift has been made from mean pole position of the several vertical sequences of traps. This indicates that a drift of the Indian landmass of the order of 24° took place during the entire period represented by the Deccan trap activity. The results of paleomagnetic studies of other sedimentary and igneous formations of Cretaceous age are reviewed in the light of recent results on Deccan traps.

1. Introduction

Mount Pavagarh (22°30′N73°30′E) in Gujrat State of India is an isolated hill composed of Deccan trap and is situated between the main Deccan trap exposure and the Kathiawar peninsula. Here nearly horizontal volcanic flows are exposed from elevation of nearly 400 ft. to about 2800 ft. above sea level. The Deccan trap itself is similar in areal extent to the Columbia River basalt in U.S.A. and Serra Geral formation in South America. The traps at present occupy an area of nearly 200,000 sq. miles extending from nearly 69.5°E to 82°E and from 16°N to 23°N. Traps originally probably covered a much larger area, of the order of 500,000 sq. miles including a large segment which now forms a part of the Arabian Sea to the west of Bombay. Paleomagnetically this formation has been studied very widely. Several investigators have contributed significantly to the paleomagnetic study of this formation. Significant contributions have been made by Clegg et al. (1956), Deutsch et al. (1959). Sahsrabudhe (1963), Bhimasankaram and Pal (1968), Athavale (1970), Pal et al. (1971), Verma and Mital (1972), Verma and Pullaiah (1972). Fig. 1 shows the location of the collection sites from which Deccan trap has been studied so far.

The early studies by Deutsch et al. (1959) and Sahsrabudhe (1963) along the west coast of India revealed a geomagnetic field reversal within the Deccan lavas which was found to be located near an elevation of about 2,000 ft. above mean sea level. The rocks above this elevation were found to be normally magnetized (declinations close to 360°) and those below this elevation were found to be reversely magnetized (declinations close to 180°). Recent studies by Bhimasankaram and Pal (1968), Pal et al. (1971) from several localities including Ellora, Aurangabad, Chandor, Ankai, Harsul and Jalna in the northwestern part of the main Deccan trap exposure and of Verma and Pullaiah (1972) as well as Athavale (1970), from Jabalpur, Dindori and Amarkantak areas, have revealed that this field reversal is located at different elevations.

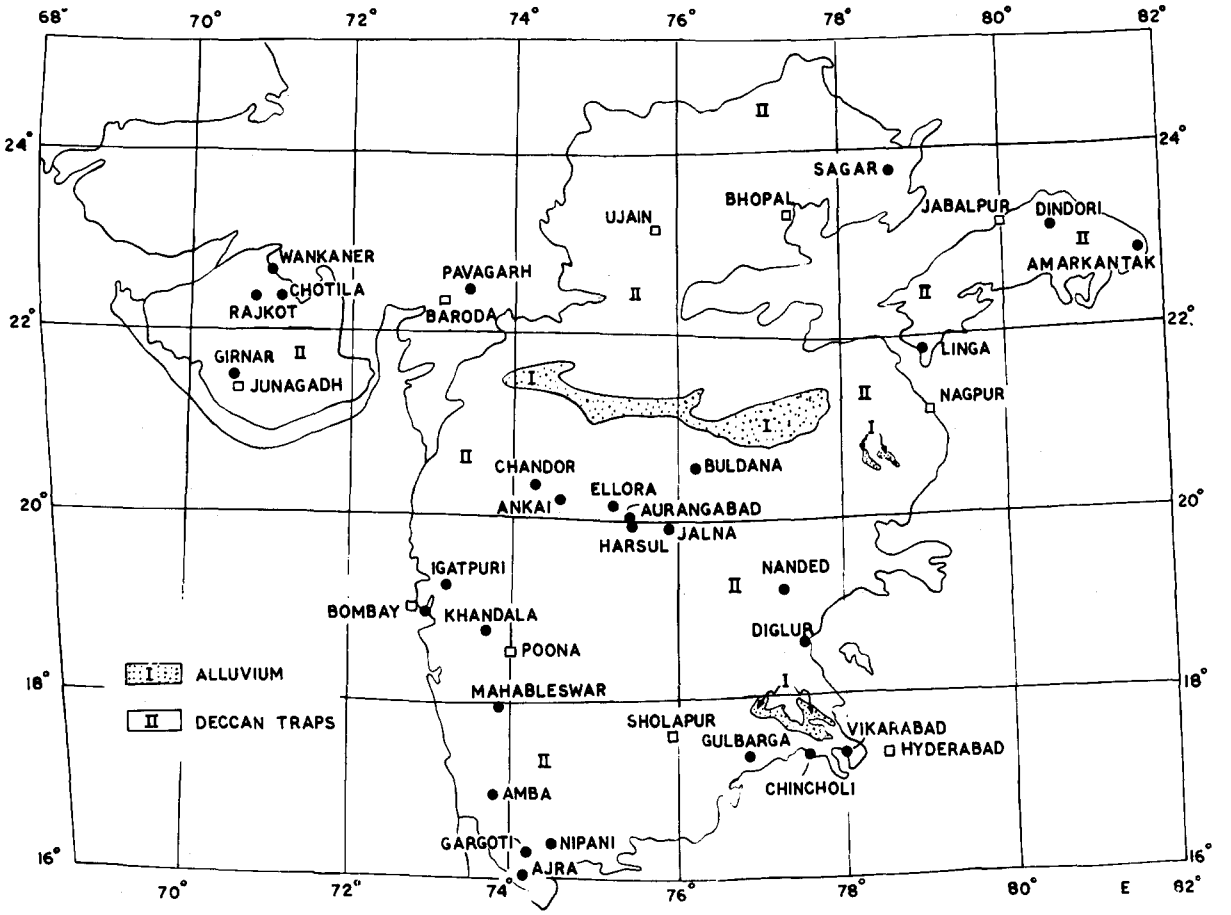


Fig. 1. Locations of the various sites from which Deccan traps have been studied so far.

Bhimasankaram and Pal (1968), Pal et al. (1971), Bindu Madhav and Pal (1972) have recently given strong evidence in support of existence of five reversals within the Deccan traps.

In this paper paleomagnetic studies of twenty flows from Mount Pavagarh are reported. The results are interpreted in terms of: (a) correlation of field reversal with other areas; (b) history of igneous activity of Deccan trap; (c) paleosecular variation; and (d) continental drift of the Indian landmass during the period of Deccan volcanic activity.

2. Geology of Deccan Plateau and Mount Pavagarh

Over most of the territory, the Deccan trap overlies the Precambrian crystalline rocks of peninsular India. In few areas these also overlie Mesozoic formations. In the Jabalpur area, the traps overlie the Lameta beds, fossil evidence for which suggests a Turonian age (Middle Cretaceous) according to Krishnan (1960, p.481). In Narmada Valley the traps are underlain by the Bagh Beds (Cenomanian–Sonomian age). For some inter-flow beds, the age of Eocene has been suggested from fossil evidence (Sahni, quoted by Krishnan, 1960, p. 484). It has been generally concluded by Oldham (in Krishnan, 1960) as well as Krishnan from geological evidence that the Deccan traps commenced eruption during the Late Cretaceous, and the activity continued intermittently well into the Eocene (Krishnan, 1960, p. 485).

Radioisotope ages have been determined for few localities in the Deccan traps. Wellman and McElhinny (1970) have reported radiometric dates of 62 ± 2 m.y.

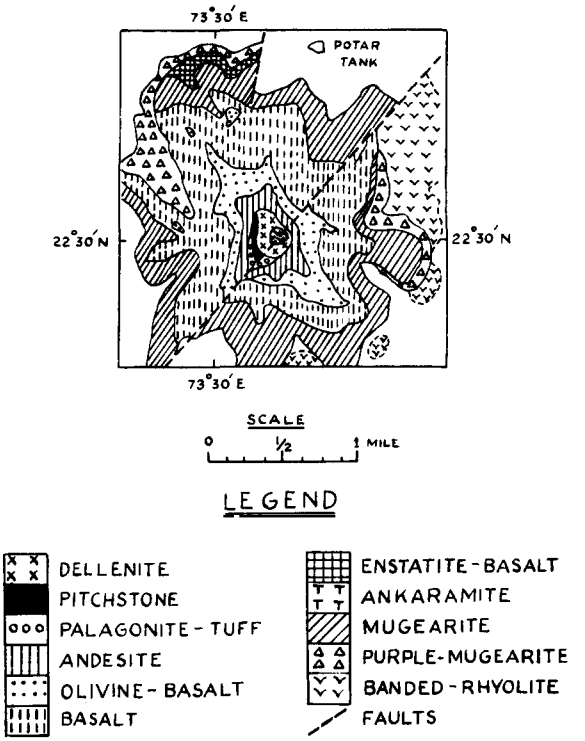


Fig. 2. Geological map of Mount Pavagarh after Sinha and Tiwary (1964).

for basal flows and 62 ± 1.2 m.y. for the uppermost flows from Mount Pavagarh. They have also obtained an age of 64.1 ± 1.4 m.y. for the youngest intrusives of Mount Girnar (21°26′N70°28′E) and 59.0 ± 1.1 m.y. for southeastern Deccan basalt (17°40′N77°36′E) near Hyderabad. Rama (1964) on the other hand has reported ages ranging from 43 ± 2 to 65 ± 5 m.y. for the top and bottom flows from Mount Pavagarh. In the area around Bombay he has dated a trachyte at 60 ± 3 m.y. and a flow and a dyke at 42 ± 6 and 45 ± 3 m.y., respectively.

Besides Deccan traps, igneous activity in the east is represented by the Rajmahal and Sylhet traps. Geologically, the Rajmahal and Sylhet traps have been considered to represent early phases of the Deccan volcanic activity (Fox, 1935). Pascoe (1964) also supports this idea saying, "Further investigations may eventually prove that the Rajmahal traps represents but the initial phase of Deccan disturbance." Between the Rajmahal hills and the Deccan traps, dykes of dolerite petrologically identical with hills and dykes of Deccan trap are intrusive into the Gondwana basins of Jharia, Raniganj, Bokaro, Karanpura and other coalfields. Athavale and Verma (1970), from studies of paleomagnetism of these dykes, have established that the igneous activity represented by these dykes is intermediate in age between those of the Rajmahal and Deccan traps.

The Rajmahal traps have now been dated at 100–105 m.y. (McDougall and McElhinny, 1970). It appears therefore that the entire igneous activity in peninsular India has lasted approximately from 100 ± 5 m.y. to 45 ± 5 m.y. (i.e., Late Cretaceous to Eocene).

Mount Pavagarh is an outlier of the Deccan formation at the northwestern part of the main Deccan trap exposures. It is surrounded on all sides by Precambrian metamorphic rocks. Fig. 2 (after Sinha and Tiwary, 1961) shows the nature of rocks comprising this exposure. The rocks comprising this hill range in composition from olivine basalt to normal basalt, mugearite, andesite, latite to pitchstone, dellinite and rhyolite. The flows over a large part of the exposure are nearly horizontal. Elevation ranges from nearly 450 ft. to nearly 2800 ft. Geologically the igneous activity represented by these flows has been considered to belong to the main Deccan Plateau Basalt Series.

3. Sampling and laboratory studies

Sampling for paleomagnetic studies was carried out from the base to the top of the hill. Five to seven hand-oriented samples for paleomagnetic studies were collected from each flow. A microaltimeter was used for determining elevations of the flows. The samples collected varied from olivine gabbro to basalt, mugearite, andesite and rhyolite. The samples were studied in the paleomagnetic laboratory of the National Geophysical Research Institute at Hyderabad. The sensitivity of the astatic magnetometer used was $2 \cdot 10^{-6}$ Oe/cm deflection.

For purposes of removing unstable components of magnetization seven to eight pilot specimens from each flow were cleaned in successively increasing alternating fields of 25, 50, 75, 100, 150 and 200 Oe (peak value) in a demagnetizing apparatus similar to that of Creer (1959). The optimum field necessary for magnetic cleaning was determined for each site,

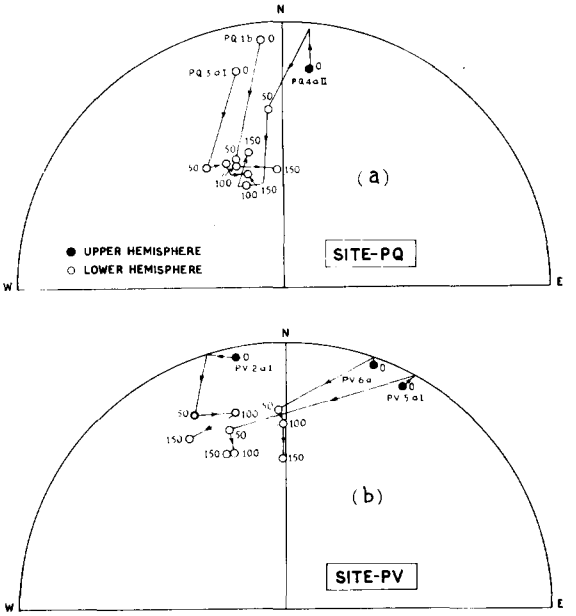


Fig. 3. Changes in remanent magnetization vectors on successive demagnetization in peak alternating fields for two sites plotted on equal-area projection. Symbols: ● = down dip; O = updip.

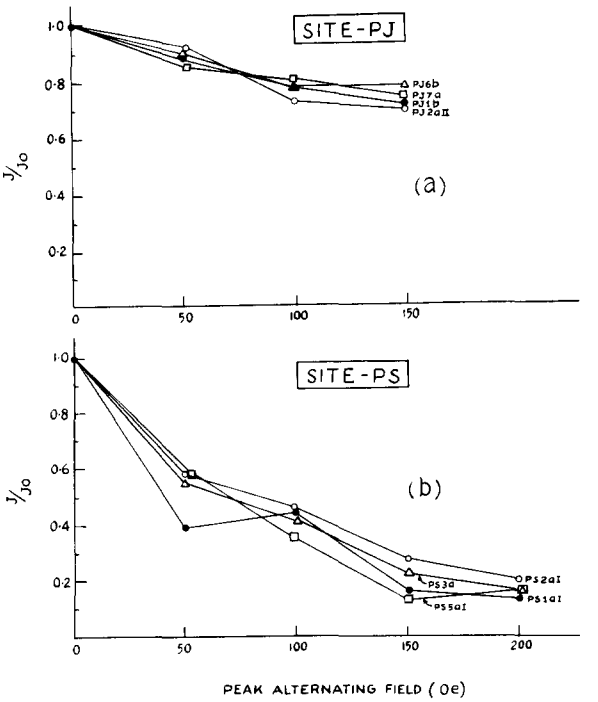


Fig. 4. Alternating field-demagnetization curves of different types: (a) those showing no appreciable change in intensity up to peak alternating fields of about 150 Oe, and (b) others showing decrease in intensity up to 75–80% of N.R.M. value. From some sites the curves showed behavior intermediate between these two types.

and remaining samples were subjected to demagnetization in that field. Samples from some of the sites could be cleaned adequately in fields upto 50 Oe while for others cleaning in fields of 150–250 Oe was necessary.

As a result of a.c.-cleaning the samples changed their directions considerably. This is shown for two typical sites in Fig. 3. Demagnetization curves, showing decrease in magnetic intensity upon magnetic cleaning for various sites were mainly of two types: those showing a decrease in magnetic intensity up to 20–25% of N.R.M. value (as shown in Fig. 4a) others showing a decrease in intensity by as much as 75–80% after treatment in 150–200 Oe peak fields (as shown in Fig. 4b). Samples from few sites showed behavior intermediate between these two types. Resultant mean directions after a.c.-cleaning are tabulated in Table I.

The physical properties which are of much-interest for these flows are: the N.R.M. intensity (J_N), susceptibility (K) and Koeningsberger ratio (Q_N). The N.R.M. intensity was measured on the astatic magnetometer while the magnetic susceptibility for each disc specimen was determined in an apparatus similar to the one constructed by Likhite and Radhakrishnamurty (1965).

It was noted that all the three parameters show a considerable variation. The maximum range in N.R.M. intensity was from $0.5 \cdot 10^{-3}$ to $7 \cdot 10^{-3}$ with several flows having values near $3 \cdot 10^{-3}$ G. The range in susceptibility was also considerable: from $0.2 \cdot 10^{-3}$ to $16 \cdot 10^{-3}$ with mean value for several flows near $5 \cdot 10^{-3}$ G/Oe. The maximum range in Q_N-ratio was from 0.5 to 45. A number of flows had Q_N-values close to 1 or less than 1, yet their precision parameter k (Table I) was quite high (mean value 94). This showed that the precision parameter depends more upon the stability of N.R.M. rather than the Q_N-ratio.

TABLE I
Paleomagnetic parameters for Mount Pavagarh

Flow No.	Elevations (ft.)	N	n	D (°)	I (°)	R	k	α_{95} (°)	V.G.P. λ_p (°N)	V.G.P. L_p (°W)	δm	δp	Treatment (A.F.D.) (Oe)	Polarity
PB	450	7	35	224	−29	6.94	100	5.3	32.5	159.0	7.2	3.2	50	intermediate
PA	560	6	23	270	−46	5.77	22	12.3	23	176.8	15.7	10.0	250	intermediate
PC	650	9	45	343	+23	8.90	80	5.2	70.7	46.7	5.5	2.9	100	normal
PD	730	7	40	295	+28	6.91	67	6.3	28.4	11.1	7.1	3.9	150	normal
PF	900	5	17	335	−43	4.97	133	5.4	36.6	78.0	6.71	4.2	50	normal
PG	950	6	22	339	−26	5.96	125	5.1	48.3	74.9	5.5	3.0	100	normal
PH	1125	5	16	327	−44	4.95	80	7.0	32.1	71.1	8.8	5.5	50	normal
PI	1180	6	19	329	−44	5.94	83	6.3	33.1	72.8	7.8	5.0	50	normal
PJ	1240	7	23	329	−19	6.96	150	4.8	45.5	60.2	5.0	2.6	50	normal
PL	1500	7	31	348	−50	6.92	75	6.1	29.8	95.4	7.6	6.3	100	normal
PM	1560	3	10	320	−21	2.99	200	5.7	39.0	52.1	6.0	3.2	50	normal
PN	1590	5	17	337	−63	4.95	80	7.0	19.8	89.3	11.0	8.7	50	normal
PO	1680	5	13	324	−43	4.94	67	7.6	31.0	68.0	9.5	5.9	50	normal
PP	1735	4	15	331	−36	3.98	150	5.7	38.9	70.6	6.6	3.9	150	normal
PQ	1825	7	30	342	−44	6.96	150	4.3	38.7	85.6	5.4	3.4	50	normal
PR	1875	5	21	344	−35	4.96	100	6.3	45.4	84.8	7.2	4.2	150	normal
PS	2135	6	21	334	−28	5.97	167	4.4	44.8	69.8	4.8	2.6	150	normal
PT	2180	6	21	335	−43	5.94	83	6.6	36.7	78.0	8.2	5.1	150	normal
PU	2465	5	22	333	−35	4.95	80	7.0	40.6	72.2	8.1	4.6	150	normal
PV	2540	6	20	344	−24	5.96	125	5.1	51.6	80.8	5.5	2.9	50	normal
PW	2600	6	28	Scattered directions										
PX	2680	7	23	Scattered directions										
Mean (16 sites) PS–PV		88	318	334	−38	15.60	38	5.43	39.22	74.39	6.76	3.99	λ_m = 25° S	

N = total number of samples; n = total number of specimens; D = mean declinations of stable remanence, east of north; I = mean inclination of stable R.M., down +ve and −ve up; R = lenth of vector sum of N unit vectors; k = precision parameter; α_{95} = radius of circle confidence at 95%-probability level; V.G.P. = Virtual Geomagnetic Pole (north); λ_p and L_p = latitude and longitude of V.G.P.: δm and δp = radii of ovals of confidence about the virtual pole position; λ_m = paleolatitude of reference town Nagpur (21°N, 79°E); A.F.D. = Alternating Field Demagnetization.

4. Results and discussion

The results of magnetically cleaned directions are given in Table I. These include mean direction for each flow as well as their corresponding Virtual Geomagnetic Poles (V.G.P.), along with other paleomagnetic parameters.

The paleomagnetic results from these flows are interpreted in terms of field reversal in the Deccan traps, paleosecular variation and continental drift.

4.1. Field reversal

Mean directions for each flow, obtained after a.c.-cleaning is plotted on polar equal-area projection in Fig. 5. Variation of declination and inclination with elevation from bottom flows to the top is shown in Fig. 6. An important feature of the result is that except for the lowermost four flows (*PB, PA, PC, PD*) all the remaining ones (*PF–PV*) show a north-northwest declination with upward dip. These directions are consistent with a paleoposition of India about 25–30° south of the equator at the time of magnetization of the flows.

A well-known feature of the paleomagnetic results on Deccan traps is a transition from southward declination and downdip to northward declination and updip indicative of a field reversal. This field reversal was first reported to be near an elevation of 2000 ft. by Deutsch et al. (1959) from their studies in the Western Ghats. Subsequently this reversal has been found to be located at different elevations at different places as shown in Fig. 7 (Bhimasankaram and Pal, 1968; Athavale, 1970; Pal et al., 1971; Verma and Pullaiah, 1972). In general, the direction of magnetization of flows with reverse polarity is found to be close to 152° ± 25° declination and 50° ± 20° downward inclination, while those of normal ones is nearly 335° ± 25° declination and 45° ± 20° upward inclination.

Intermediate directions have been observed only at a few places such as near Dindori (Verma and Pullaiah, 1972), Mahableshwar (Sahsrabudhe, 1963), Chandor and Ellora (Bhimasankaram and Pal, 1968). The directions of magnetization of the lowermost two flows at Mount Pavagarh seem to be intermediate between those of reverse and normal polarity of the Deccan traps and hence could possibly represent this transition period. The direction of two flows above

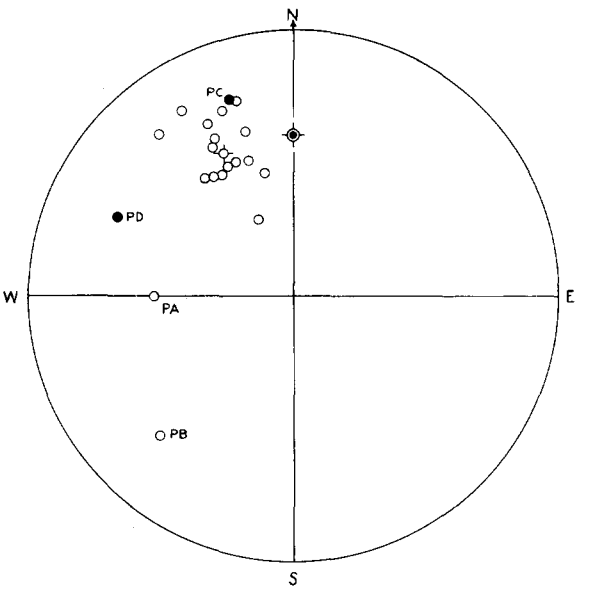


Fig. 5. Mean direction for each site after a.c.-cleaning plotted on polar equal area projections. Symbols: • = down dip; O = updip, ✧ = mean direction of flows; and ✧ = direction of present dipole field at site.

these, namely *PC* and *PD* are seen to be discordant. Such directions have been observed in Deccan traps in Amarkantak by Athavale (1970), in Sagar by Pal et al. (1971) as well as in Aurangabad areas (Athavalie et al., 1972).

One explanation for such discordant directions is that the landmass of India was in the northern hemi-

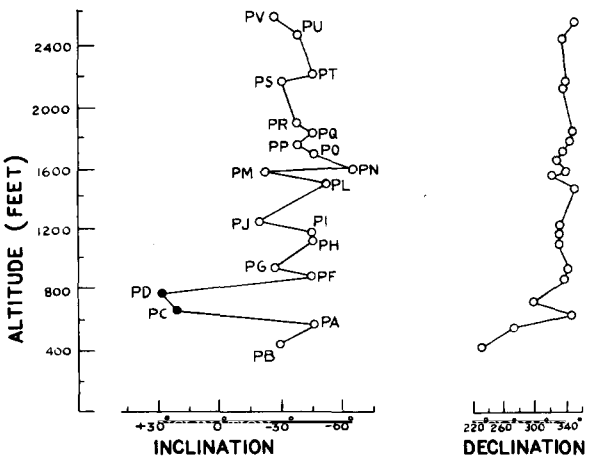


Fig. 6. Variation in declination and inclination with elevation from bottom to top flows at Mount Pavagarh.

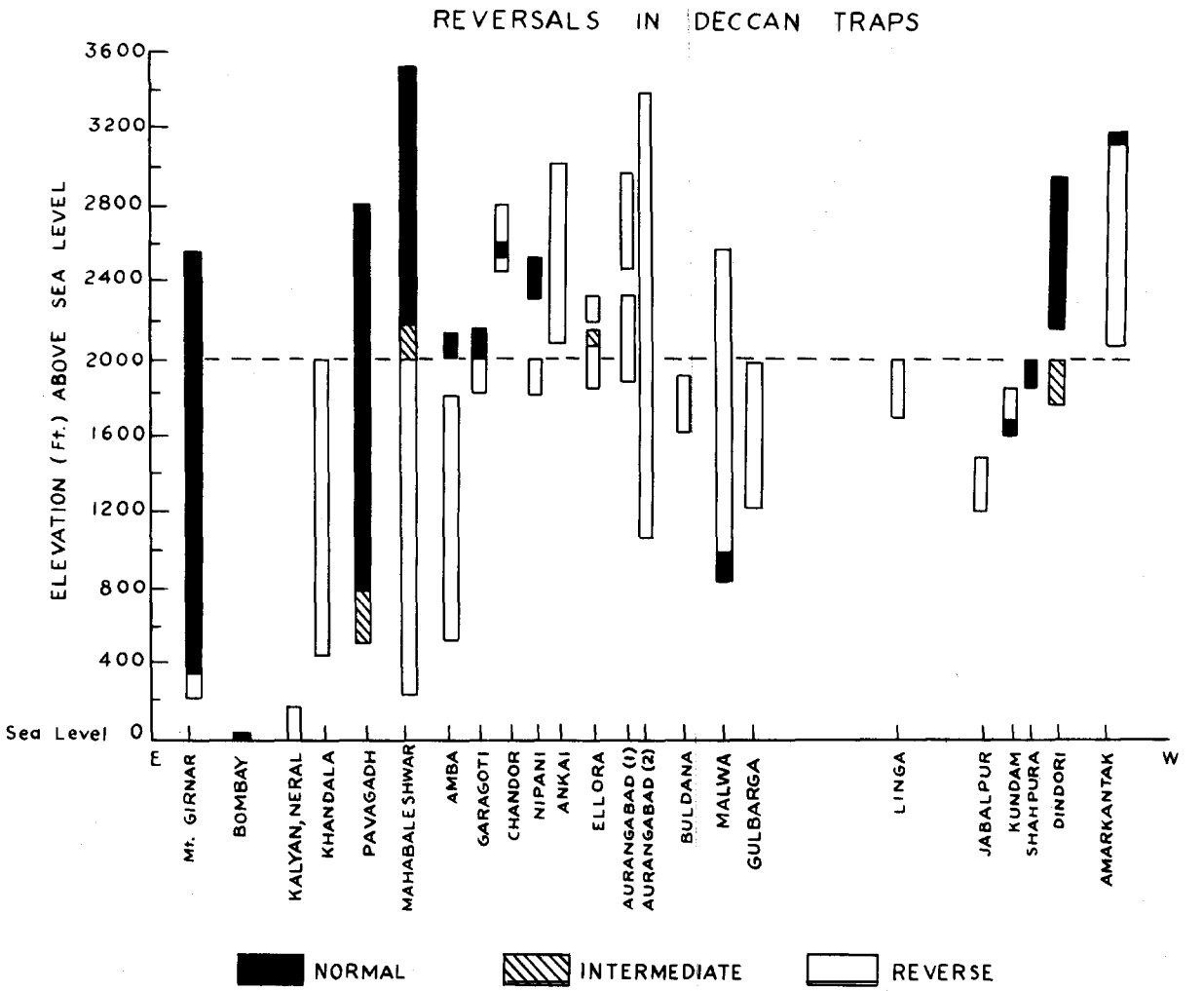


Fig. 7. Nature of magnetic polarity on Deccan Lava flows from various localities shown in Fig. 1. The figure clearly shows the predominance of one normal and one reverse polarity in these lavas.

sphere at the time of eruption of these flows. However, this possibility is completely ruled out, as all the upper flows at Mt. Pavagarh and the directions of large numbers of flows from Deccan traps of various localities in India indicate that the landmass of India was in the southern hemisphere during this period. It may be noted that the N.R.M. intensities and Q_N-ratios of flows *PC* and *PD* are not in any way abnormal and hence their anomalous directions could not be ascribed to any lightning effect. It seems most likely therefore, that the directions of these flows are representative of a period of instability of the geomagnetic field associated with field reversal. This interpretation is strengthened by the observation that the underlying two flows show intermediate directions. It is theoretically presumed (Elsasser, 1956) that during a field reversal the strength of the main dipole reduces to a minimum and increases again in the opposite direction. Under such circumstances the nondipole field is likely to become considerably predominant, and hence temporarily a resultant field direction far removed from that due to a dipole field is possible.

Few words may be said regarding the nature of field reversals in the Deccan traps. As observed by several workers, at most of the locations normal rocks have been found to be overlying those with

reverse polarity (Fig. 7). Thus, there appear to be two long periods of time during which the field had a reverse and (later) a normal polarity. Pal et al. (1971) and Bindu Madhav and Pal (1972) have given strong evidence in support of five field reversals during the Deccan trap period from their study of a vertical sequence of flows at Malwa and Sanwad. Similar results have also been observed from Dindori region near Jabalpur (G. Pullaiah, personal communication, 1970). It therefore seems that there have been two long periods of normal and reverse polarity indicating one major field reversal and few other reversals lasting for shorter periods of duration during the entire Deccan trap period.

From all available evidence (see Fig. 7) it appears that the long period of normal polarity as observed at Mt. Pavagarh is probably the same as that of Mt. Girnar (Verma and Mital, 1972) or Mahableshwar (Sahsrabudhe, 1964) and Jabalpur (Verma and Pullaiah, 1972).

4.2. Paleosecular variation

Considering that each mean flow direction represents a spot reading of the paleomagnetic field, we can get some idea of the nature of the geomagnetic field and its secular variation during the period of its singular polarity from the scatter of the observed flow directions or that of their corresponding V.G.P. (Cox and Doell, 1964). We have adopted the latter procedure for our study. The V.G.P. corresponding to each flow from Mt. Pavagarh are given in Table I.

The total angular dispersion (S_T) observed for all the flows consists of two parts: between-site dispersion (S_B) mainly due to ancient geomagnetic secular variation, and within-site dispersion (S_W). The two are related as follows (Doell, 1970):

$$S_T^2 = S_B^2 + (S_W^2/\bar{N})$$

where $\bar{N}$ is the average number of samples per flow; S_W is taken to be the mean of individual flow angular standard deviations; and:

$$S_T = (N-1)^{-\frac{1}{2}} \left(\sum_{i=1}^{N} \delta_i \right)^{\frac{1}{2}}$$

where δ_i is the angular deviation of the i-th pole from the mean pole for the whole sequence. In the case of 16 flows with normal polarity for the Mt. Pavagarh, $S_T = 12.2°$, $S_W^2/N = 11.6°$. The between-site dispersion also includes a small component due to local magnetic anomalies that existed at the sampling site at the time of cooling of the lavas. On account of several unknown factors regarding the history of eruption of traps and pre-Deccan trap topography, no correction has been made for this factor. This is, however, estimated to be very small, of the order of 1° or less. Limits of 95% confidence for the standard deviation were calculated according to the method suggested by Cox (1969). Standard deviations for pole positions for 16 normal flows from Mt. Pavagarh, are plotted vs. paleolatitude of sites in Fig. 8. Similar calculations were made for three other vertical sequences of Deccan traps which have been studied thoroughly. Various flows in each of these sequences have been sampled adequately and stability tests have also been applied.

These are 14 flows with normal polarity from Mt. Girnar in Gujrat ranging in elevation from 470 ft. to 2470 ft. (Verma and Mital, 1972), 25 flows with reverse polarity from Aurangabad ranging in elevation from 1140 ft. to 3000 ft. (Athavale and Anjaneyulu, 1972) and 15 flows from Khandala with reverse polarity ranging in elevation from 400 ft. to 2000 ft. (Sahsrabudhe, 1963). The standard deviation of angular dispersion for the normal sequence of Mt. Girnar and reverse sequence of Khandala and Aurangabad along with 95%-confidence limits are also shown in Fig. 8. Paleomagnetic parameters for these sequences are given in Table II.

It can be seen that the angular dispersion for flows from Mt. Girnar and Mt. Pavagarh is much smaller than that of Khandala and Aurangabad. The significance of these results can be better understood by comparing these with theoretical models of angular dispersion for a magnetic field in which we have dipole wobble only, and another one in which nondipole field similar to the present one is superimposed on a dipole field with a wobble of about 11° as for Bruhnes epoch (Doell, 1970). We find that the angular dispersion represented by flows from Mt. Pavagarh can best be explained by having some non-dipole components in addition to the wobble of the main dipole field. The angular dispersions represented by Khandala and Aurangabad flows can be attributed to larger contributions due to the nondipole field. Overall, the results indicate that the magnitude of

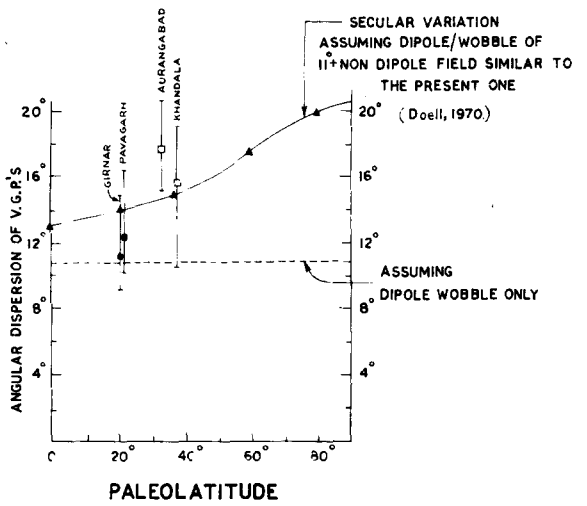


Fig. 8. Angular standard deviation of Virtual Geomagnetic Poles (V.G.P.) for various vertical sequences of Deccan lavas (discussed in text) plotted vs. paleolatitude of sites.

secular variation has varied considerably during the period of Deccan trap activity.

4.3. Continental drift

Various investigators have made estimates of the drift of the Indian landmass during the period of eruption of the Deccan Basalt. Deutsch et al. (1959) gave an estimate of drift of about 50° in latitude during the past 70 m.y. This was, however, based upon their results of paleomagnetic studies of several flows from the Western Ghats including localities of Khandala, Linga, Nipani, Amba, Igatpuri, Kambatki and Pavagarh. The large estimate of continental drift was primarily based upon the difference in the pole positions between that of Khandala and the uppermost flows of Mt. Pavagarh. Our results for the latter flows as given in this paper are considerably different from theirs. In our studies the uppermost rhyolite flows were found to be unstable after applying a.c.-demagnetization tests. The rest of the acidic as well as basic flow were found to have declination ranging from 326° to 344° and with upward inclinations ranging from 16° to 63°. The mean direction of 16 flows with normal magnetization was found to be as follows: declination = 334° and inclination = −38°. This is different from the mean direction of basic flows from Pavagarh as reported by Deutsch et al. (1959): $D = 351°$ and $I = -16°$.

McElhinny (1968) analyzed the data on the Deccan traps from various localities and suggested that only few degrees of drift took place during the entire period of Deccan trap activity, which may have lasted for a period of about 5 m.y.

Recent paleomagnetic data from different localities including Jabalpur, Aurangabad, Western Ghats, M. Pavagarh and Mt. Girnar has thrown a new light on the results from Deccan traps.

We have obtained the mean pole positions for several vertical sequences of normal as well as reverse flows studied from one locality. These include 8 flows from Jabalpur (Verma and Pullaiah, 1972), 14 flows from Mt. Girnar (Verma and Mital, 1972), 16 flows from Pavagarh (present study), all normal polarity, and 90 flows from Western Ghats near Poona (Wensink and Klootwijk, 1971), 25 flows from Aurangabad (Athavale and Anjaneyulu, 1972) and 15 flows from Khandala (Sahsrabudhe, 1963), all with

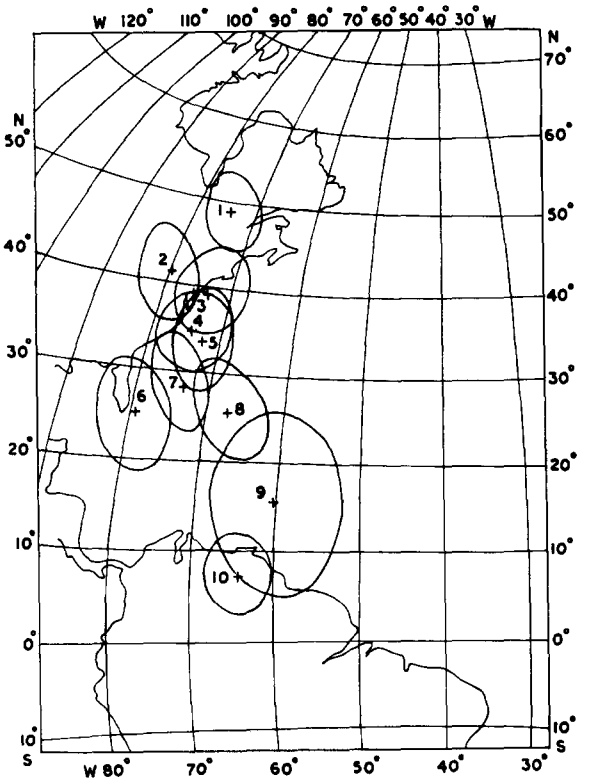


Fig. 9. Mean pole positions for six vertical sequences of Deccan traps and other sedimentary and igneous formations of Cretaceous age as listed in Table II.

TABLE II
Paleomagnetic data for six vertical sequences of flows from Deccan traps (D.T. 1–6) and other sedimentary and igneous formation

No.	Locality	Elevation (ft.)	Age	Number of flows or sites	D (°)	I (°)	R	k
1	D.T. Jabalpur (Dindori) 23°30'N80°57'E	2150–2900	–	8	343	–28	7.92	87.5
2	D.T. Mount Girnar 21°30'N70°30'E	470–2470	Paleocene	14	336	–38	13.82	72
3	D.T. Mount Pavagarh 22°30'N73°30'E	900–2540	-do-	16	334	–38	15.60	38
4	D.T. Western Ghats 17°51'N76°47'E	–	–	90	151.7	+49.6	–	403
5	D.T. Aurangabad 19°51'N76°16'E	1140–3000	–	25	150	+48	–	–
6	D.T. Khandala 18°45'N73°22'E	400–2000	–	15	147	+58	14.7	47
7	Tirupati Sandstones 16°48'N81°10'E	–	M. Cretaceous	3	153	+56	–	–
8	Satyavedu Sandstones 13°30'N80°E	–	-do-	1	321	–58	–	–
9	Sylhet trap 25°N91°E	–	Jurassic, Cretaceous?	–	332	–59	–	–
10	Rajmahal traps 24°15'N87°45'E	–	100–105 m.y.	48	316	–65	–	–

reverse polarity. The data for mean directions for these flows, their range of elevations, pole positions, radii of circle of confidence, etc. are given in Table II. The pole positions are plotted in Fig. 9 along with limits of their probability error at 95%-confidence level. The exact time span represented by the igneous activity in these areas is difficult to estimate. However, if the average time between successive flows is of the order of 500–1000 years, the secular variation should be averaged out in almost each of the vertical sequences studied.

Paleomagnetic data from other igneous and sedimentary formations of Cretaceous age are listed in Table II and their pole positions are also plotted in Fig. 9. From the paleomagnetic and the geological data (discussed earlier) the following observations can be made:

(1) The pole positions for Mt. Girnar (no. 2) and Mt. Pavagarh (no. 3) both of normal polarity are close to each other, suggesting that the igneous activity represented by these areas is nearly contemporaneous. Radiometrically also their dates have been found to be fairly close to each other: 61–64 m.y.

(2) The paleomagnetic results from these formations, represented by poles *1–10* can be interpreted in terms of drift of the Indian landmass from the southern hemisphere towards the northern hemisphere. For purposes of estimating the amount of drift, the paleolatitude of a centrally located city, Nagpur (latitude 21°N), has been calculated and is also given in Table II. The difference in the paleolatitude of Nagpur corresponding to paleomagnetic results from Mt. Girnar and Rajmahal traps (poles *2–10*) is 19°. The minimum difference in their radiometrically determined ages is 40 m.y. This implies a drift rate for the Indian landmass of the order of 5.2 cm/year between the time of eruption of these volcanic series.

(3) Out of the six pole positions (*1–6*) given for

of Cretaceous age from India (symbols as in Table I)

α_{95} (°)	V.G.P.		λ_m (°S)	δp (°)	δm (°)	Polarity	Reference
	λ_p (°N)	L_p (°W)					
5.0	48	74	15	3.0	6.0	normal	Verma and Pullaiah, 1972
4.4	41.2	79.3	25	3.0	5.2	normal	Verma and Mital, 1972
5.7	39.4	74.8	25	3.9	6.7	normal	present study
3.8	34.5	76.4	30	3.4	5.1	reverse	Wensink and Klootwijk, 1971
5.5	33	73	30	4.7	7.2	reverse	Athavale and Anjaneyulu, 1972
5.0	25	79	39	5.0	7.0	reverse	Sahsrabudhe, 1963
4	28	73	34	2.5	7.2	normal and reverse	Verma and Pullaiah, 1967
4	26	67	33	4.3	5.9	normal	Mital et al., 1970
7	16	60	35.5	7.8	10.5	normal	Athavale et al., 1963
2.5	7.5	63.5	44	3.0	3.5	normal	Klootwijk, 1971

various sequences of the Deccan traps, no. *6* is the southernmost and no. *1* the northernmost. Difference in the paleolatitude of Nagpur corresponding to these poles is 24°. This suggests that a significant amount of drift of the landmass took place during the time of eruption of the main Deccan basalts.

(4) The pole positions corresponding to Tirupati and Satyavedu formations (no. *7* and *8*) are close to that of Khandala. On the basis of fossil evidence these formations are considered to be of Middle Cretaceous age. However, their pole positions indicate that the magnetizations of these formations could be of younger age, i.e., lower Deccan traps or Late Cretaceous.

(5) The pole position for Khandala (no. *6*) is intermediate between that of Girnar and Rajmahal of 64 ± 1.5 and 100 ± 5 m.y. ages, respectively. This suggests that the igneous activity represented by these flows could be of intermediate age between these two or approximately 80 ± 10 m.y.

By similar reasoning the Sylhet traps appear to be younger than the Rajmahal traps.

The paleomagnetic results from the Cretaceous–Paleocene or Eocene igneous and sedimentary formations from India indicate that the eruptions of the Deccan and Rajmahal lavas was a major event in the history of the Indian subcontinent and was associated with appreciable drift of the Indian landmass. Even when these eruptions had ceased, the landmass was in the southern hemisphere as indicated by paleolatitude of Nagpur for various flow sequences.

A few words may be said regarding the correlation of these results with those obtained from sea-floor spreading in the Indian Ocean. Le Pichon and Heirtzler (1968) have suggested that Africa began to be separated from India, Australia and Antarctica along the southwestern branch of the Indian Ocean Ridge about 140 m.y. ago. This time of breakup of Gondwanaland is consistent with the present paleo-

magnetic data. They have further suggested that the opening between Africa in the northwest side and India, Australia and Antarctica on the southeast side of the ridge was completed during Albian times (100 m.y. ago). This was the approximate time of commencement of the volcanic activity represented by the Rajmahal trap in the east. As the drift of the Indian landmass continued northward at a relatively fast rate (about 5 cm/year), the magnitude of volcanic eruptions increased resulting in huge outpouring of the Deccan lavas on the subcontinent as well as over the area now forming the continental shelf off the west coast. The northward drift of the Indian subcontinent appears to have continued till the Miocene when the major episode in the uplift of the Himalayas took place.

We can therefore say with some confidence that the history of the volcanic eruptions in the subcontinent is intimately associated with the breaking up of Gondwanaland, formation of the Indian Ocean and subsequently with the birth of the Himalayas.

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